Memory	Minimum 4 GB RAM recommended for smooth browser and backend execution.
Operating System	Windows 10/11 or any OS capable of running Python and Flask.
Backend	Python, Flask, SQLite, requests and openpyxl.
Frontend	HTML, CSS, JavaScript, Leaflet, Chart.js and OpenStreetMap tiles.
Data Files	Pingstation captured log files, SQLite database, CSV exports and ICAO Excel workbooks.

2.6 Feasibility Study

Technical feasibility is supported by the implemented operating limits, not only by tool availability. The dashboard requests aircraft data every 1 s and status every 4 s. Live aircraft are removed after 45 s without a frame; replay records use a 12 s stale limit. A trajectory point is stored when it is the first sample, at least 5 s has elapsed, movement is at least 30 m, altitude changes by 20 m, heading changes by 5 degrees, or speed changes by 2 m/s. Prediction uses the latest 4-20 samples and is limited to feed gaps of at most 180 s; if the linear fit is unavailable, constant-speed/heading kinematics is used. SQLite runs locally in WAL mode, so no separate database server is required and a 4 GB system is sufficient for the project workload. The main technical limits are map-tile/network availability, receiver data quality, and reduced accuracy during sharp turns or long outages. The system is suitable for academic monitoring, not certified air-traffic control.

2.7 Product Functions

The product functions describe the work expected from the tracker from the user's point of view. They are intentionally written in implementation-neutral language so that they can be used for verification as well as documentation.

- Acquire ADS-B related data from the configured source.
- Decode raw frames into structured aircraft records wherever field information is available.
- Filter invalid or unusable location records before visualization.
- Provide a browser-based display of live aircraft locations.
- Maintain recent movement history for selected aircraft.
- Persist trajectory data for future study and prediction experiments.
- Export trajectory and ICAO-wise records in widely readable formats.
- Provide status information about logging and export configuration.

2.8 Actors

Table 7: Actors

Actor	Role in the System